Which spaceflight biology hardware can be modelled? A metadata audit of NASA's hardware catalogues and three-dimensional transient CFD of specimen environments

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Abstract

Spaceflight biology results are interpreted against the hardware the specimens lived in, yet the physical environment inside that hardware – airflow, gas composition, temperature, humidity, light and nutrient delivery – is rarely documented well enough to be modelled. We assembled the three public NASA hardware sources (the SPLASH instrument catalogue, the OSDR hardware registry and the experiment-platform field of every OSDR study) into 328 hardware entities, of which 164 can carry space-biology payloads and 112 house live specimens in flight. We defined a 15-item minimum-information checklist for computational fluid dynamics (CFD) of a specimen environment, scored every specimen-housing item automatically (validated against two blind raters; weighted $\kappa = 0.89$ between raters, scorer precision 1.00 and recall 0.69) and found that public metadata support a CFD-ready description for 1 of 112 items; SPLASH alone supports none. Airflow rate, specimen metabolic load and heat load are the least documented inputs. Documentation is better for hardware that has been used in OSDR studies (Spearman $\rho = 0.52$) and for ISS-era hardware, and worst for cell and microbe culture hardware. Curation also exposed catalogue defects, including a block of 27 SPLASH records whose descriptive fields are displaced from neighbouring rows. Targeted literature curation, with every value tied to a verbatim source quotation, raised a subset of heavily used hardware to CFD readiness; for these we built three-dimensional transient OpenFOAM models in flight ($0\,g$) and ground-control ($1\,g$) conditions, with engineering drawings and four-dimensional field records. All code, data snapshots and curated specifications are released for reuse.

1 Introduction

Every spaceflight biology experiment is also an experiment on its hardware. Sealed canisters, fan-ventilated growth chambers, rodent habitats and culture cassettes impose their own gas, temperature and humidity fields on the specimens, and in microgravity the buoyant convection that mixes those fields on Earth is absent. When flight and ground-control specimens differ, part of the difference can therefore arise from the hardware environment rather than from gravity itself. NASA's Open Science Data Repository (OSDR) now holds hundreds of spaceflight studies [1, 2], and the OSDR also publishes two hardware resources: the SPLASH instrument catalogue [3] and a hardware registry linked to payloads and studies [4]. Whether these resources describe the hardware well enough to reconstruct the specimen environment has not been assessed.

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Here we (i) build a crosswalk across the three public hardware sources and count the hardware able to carry space-biology payloads, where it flies and how it feeds its specimens; (ii) define a minimum information checklist for CFD of a specimen environment and measure, item by item, how much of it the public record supplies, validating the automated scoring against blind human-style rating; (iii) run a meta-analysis of which hardware is under-documented; and (iv) for hardware whose parameters can be sourced, build verified three-dimensional transient CFD models in $0\,g$ and $1\,g$, with orthographic drawings and time-resolved field records. Throughout, every number either comes from a cited source, is computed from cited values, or is declared as a modelling assumption.

2 Methods

2.1 Data sources and snapshot

On 2026-10-12 we downloaded, without authentication, the full SPLASH catalogue (`splash-api/public/instru` 147 records, 62 fields), the OSDR hardware registry (`geode-py/ws/api/hardware`, 157 records, with every linked payload and experiment record), and the `experiment platform`, project identifier and organism of every OSDR dataset (641 datasets) together with each dataset's metadata, including protocol descriptions. Raw responses are archived with SHA-256 checksums; every downstream stage reads only this snapshot.

2.2 Catalogue integrity

Because SPLASH has no structured links, we tested each record's internal consistency before using it. For each descriptive field we computed TF–IDF cosine similarity between the field and the identity text (name, alternate names and background) of every record; a field that shared almost nothing with its own record but matched a record within three rows was flagged as displaced, the signature of a row-offset error. Organism classes named in the specimen field were also compared with those named in the record's own background. Every flag was reviewed by hand and given a verdict with a reason and, where possible, independent evidence from the OSDR registry (Supplementary Table S1).

2.3 Crosswalk

SPLASH and registry records were joined into hardware entities by exact match of normalised aliases and by acronyms unambiguous across both sources; joins that would merge two SPLASH records were refused. Study platform strings were split into components and mapped many-to-many onto entities (exact alias, unambiguous acronym or containment of a specific alias), so that a string naming two devices links to both without merging them. Fuzzy matches were never applied automatically: they were written out as proposals and accepted or rejected in a curated decision file, each with a reason.

2.4 Classification

Each entity was assigned a role (specimen habitat, hosting facility, processing/fixation, analysis instrument, sensor, exposure facility, ground facility or non-biological) from the SPLASH category

where present, otherwise from keywords in its name and registry description, with curated corrections. Organism classes were taken from hardware-specific text and from the organisms of linked OSDR studies. Flight platform was taken from the missions of the hardware's registry payloads. Nutrient and water delivery modes were assigned only when a verbatim quotation supported them and the mode was compatible with an organism the hardware carries.

2.5 Minimum information for CFD (MI-CFD-HW) and its validation

We defined 15 items needed to model a specimen environment (Supplementary Table S2), six of them critical: enclosure dimensions, specimen-container dimensions, ventilation mode, airflow rate, specimen temperature and specimen type. Each item scores 0 (absent), 0.5 (qualitative only) or 1 (quantitative or explicit), with the quotation that earned the score; laboratory preparation, storage and analysis sentences are excluded. A hardware item is Tier A (CFD-ready) when every critical item scores 1, Tier B when every critical item scores at least 0.5, and Tier C otherwise. Two blind raters scored a seeded random sample of 12 specimen-housing entities (180 cells) from the raw text; disagreements were adjudicated with written reasons, and the automated scorer was evaluated against the adjudicated gold standard without further tuning.

2.6 Meta-analysis

Completeness (mean item score) was compared across hardware class, flight platform, era, SPLASH status and catalogue coverage with Kruskal–Wallis tests and pairwise Mann–Whitney tests with Benjamini–Hochberg correction, and related to the number of OSDR studies using the hardware by Spearman correlation.

2.7 Literature curation

For specimen-housing hardware with OSDR studies, parameters were sought in peer-reviewed papers, NASA technical reports, mission documentation and the snapshot itself. Each parameter record carries its value, unit, source, URL and a verbatim quotation of at most 40 words; values were never transferred between hardware items or filled with typical values, and unresolved parameters are left empty with a note of where they were sought. Conflicting sources are recorded.

2.8 CFD models

For Tier A hardware we generated geometry from parametric primitives in which every dimension is either a reference to a sourced parameter or a declared assumption, and produced orthographic drawings from the same primitives. Meshes were built with `snappyHexMesh` and solved with `buoyantBoussinesqPimpleFoam` in OpenFOAM v2606 [5, 6] (laminar, second-order backward time integration). Air properties were computed with CoolProp [7–9]; CO_2 and water-vapour diffusivities are those tabulated by Monteith and Unsworth [10]. CO_2 (ppm) and water vapour ($\mathrm{mmol\,mol^{-1}}$) were transported as passive scalars with specimen sources in cell zones; heat sources enter the energy equation. Each case was run with gravity off (flight) and on (ground control).

Verification. The solver configuration was verified on the differentially heated square cavity against the reference Nusselt numbers of de Vahl Davis [11] and Hortmann et al. [12] (as tabulated

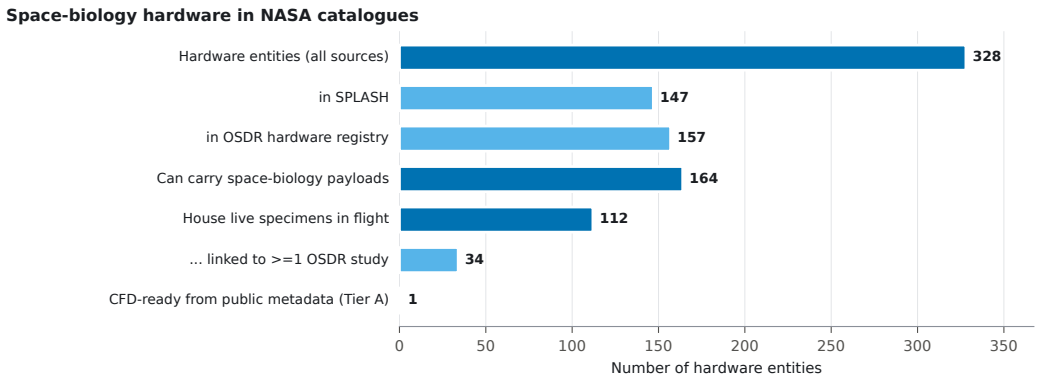


Figure 1: Space-biology hardware in the three NASA sources, from all entities to those whose public metadata support a CFD model.

by Gjesdal et al. [13]) at three Rayleigh numbers on three meshes (mean Nu 4.53, 4.52 and 4.52 at $Ra = 10^5$ versus 4.522), on one-dimensional diffusion against the analytic solution, and by closing the transient CO_2 balance (source = outflow + accumulation) in every hardware run. Discretisation uncertainty was estimated with the grid convergence index [14].

3 Results

3.1 The hardware catalogue

The crosswalk produced 328 entities (147 in SPLASH, 157 in the registry); 32 study platform components named a vehicle, location, ground analogue or generic item rather than hardware. 164 entities can carry space-biology payloads and 112 house live specimens during flight (Fig. 1). Only 34 of these are linked to at least one OSDR study. Mission records place bio-capable hardware on the Space Shuttle (38), on free-flying biosatellites (19) and on the ISS (18). Nutrient delivery was evidenced for fewer than half of the habitats (60 with no evidence); where evidenced, animal food and water (16), culture medium (21) and agar (11) were the commonest modes.

3.2 Catalogue defects

27 SPLASH records (#121–#147) carry descriptive fields displaced from neighbouring rows: 29 fields were confirmed as misattributed on review, for example a Plant Growth Unit record whose specimen field describes an RNA thermal cycler, and a BRIC-Opti canister record carrying the crop list and CO_2, humidity and temperature control ranges of the Advanced Plant Habitat. The `sources`, `schematics` and `study_links` fields are empty in every record. 310 of 641 OSDR datasets have no experiment-platform entry; the remainder use 116 free-text strings, with 9 spellings of Veggie alone.

3.3 How much of a CFD model the public record supplies

Specimen type is almost always stated (98.21% quantitative), but airflow rate (3.57%), specimen metabolic load (0.00%) and heat load (1.79%) are almost never given (Fig. 2). Median completeness was 0.30 using all public sources and 0.07 using SPLASH alone. 1 of 112 specimen-housing items

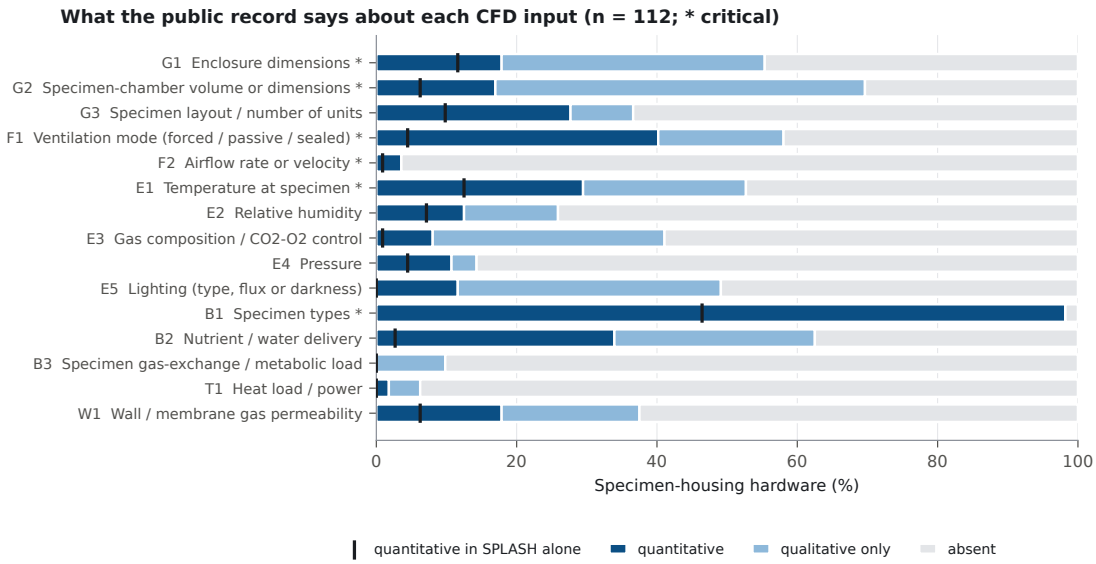


Figure 2: Coverage of each MI-CFD-HW item across specimen-housing hardware. Dark: quantitative; light: qualitative only; grey: absent. Ticks: quantitative in SPLASH alone.

reached Tier A from public metadata and 0 from SPLASH alone; waiving the airflow requirement for sealed or passive hardware raised this to 2. Blind raters agreed on 90% of 180 cells (weighted $\kappa = 0.89$). Against the adjudicated gold standard the automated scorer never credited absent information (precision 1.00) but missed some (recall 0.69; mean completeness 0.36 versus 0.53 gold), so automated completeness is a conservative lower bound; no sampled entity was Tier A on the gold scores either.

3.4 Which hardware is under-documented

Completeness differed by hardware class (Kruskal–Wallis $p = 0.00$), flight platform ($p = 1.1 \times 10^{-5}$) and catalogue coverage ($p = 9.7 \times 10^{-7}$), but not by era ($p = 0.46$) or SPLASH status ($p = 0.16$) (Fig. 3). Cell and microbe culture hardware was least documented and ISS hardware best. Hardware used in OSDR studies was better documented than unused hardware ($p = 1.8 \times 10^{-7}$; $\rho = 0.52$ with the number of studies, Fig. 4), but several used items remain below the median.

3.5 CFD of curated hardware

TODO: curated tiers, verification table, per-hardware environment cards and $0\,g/1\,g$ contrasts.

4 Discussion

TODO: discussion after CFD results.

Data and code availability

Code, the 2026-10-12 source snapshot, curated hardware specifications, model definitions, reduced CFD outputs and this manuscript's sources are archived on Zenodo (DOI to be assigned on deposit)

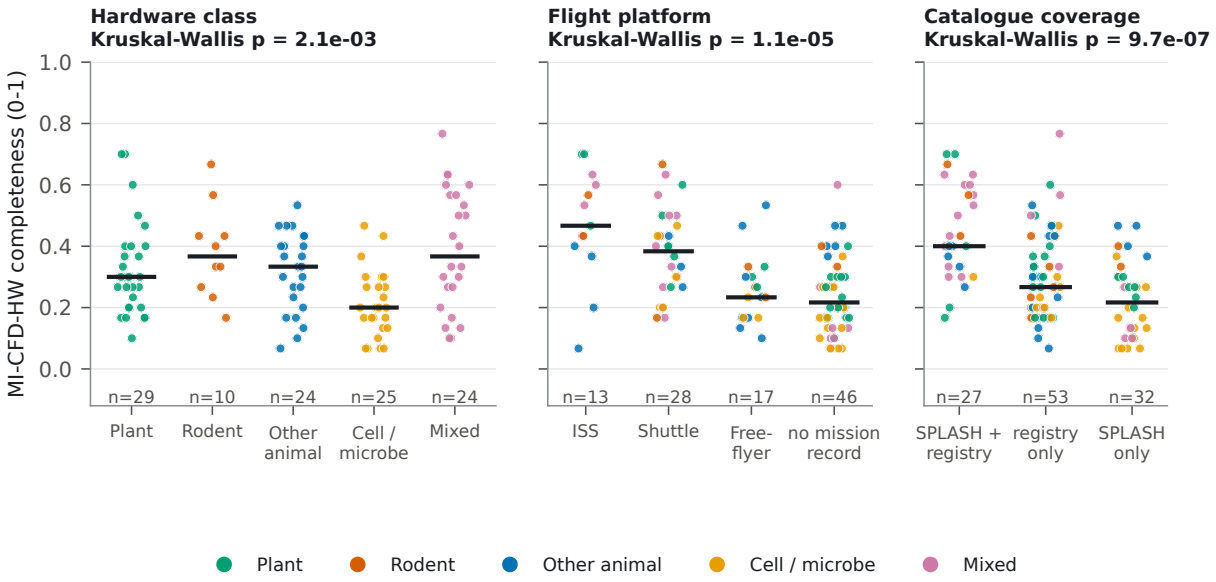


Figure 3: MI-CFD-HW completeness by hardware class, flight platform and catalogue coverage (bars: medians).

under the MIT licence (code) and CC-BY-4.0 (data and figures). SPLASH data were obtained from NASA's Space Life Sciences Archive of Scientific Hardware [3].

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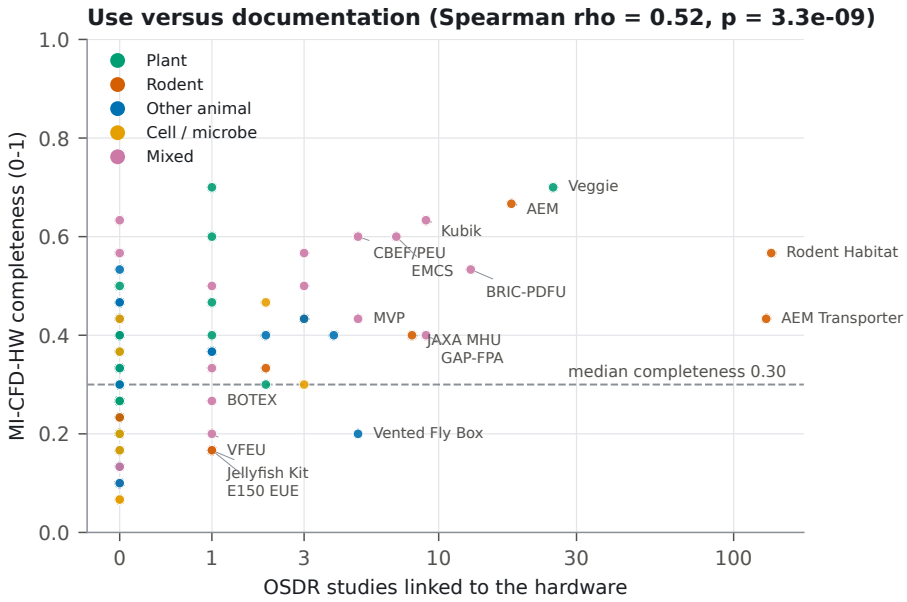


Figure 4: Documentation versus scientific use. Points: specimen-housing hardware.

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